THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · EASY LEVEL CHAPTER TEST SET 1

Chapter 8: **Heredity** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Genetics is the study of 1

(a) digestion **(b)** animal fossils
(c) heredity and variation **(d)** plants only

2. Number of chromosomes in a human body cell is 1

(a) 23 **(b)** 23 pairs plus 1 **(c)** 44 **(d)** 46

3. Which is an inherited trait? 1

(a) A broken arm **(b)** Colour of the eyes
(c) Knowing how to swim **(d)** A tan

4. Mendel worked on 1

(a) mango **(b)** wheat **(c)** maize **(d)** garden pea

5. Sex chromosomes in a human male are 1

(a) XY **(b)** XO **(c)** XX **(d)** YY

6. Which is an acquired trait? 1

(a) Blood group **(b)** Height potential
(c) A scar from an accident **(d)** Hair colour

7. A character such as tallness or dwarfness is called a 1

(a) gamete **(b)** gene **(c)** trait **(d)** cell

8. Sex chromosomes in a human female are 1

(a) XY **(b)** XYY **(c)** XX **(d)** YY

9. Acquired traits are 1

(a) present at birth **(b)** coded in DNA
(c) always passed on **(d)** not passed to the next generation

10. The part of the cell that contains genes is the 1

(a) nucleus **(b)** cell wall **(c)** vacuole **(d)** ribosome

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Who is the Father of Genetics? 2

12. What are chromosomes? 2

13. What is the F2 phenotypic ratio of a monohybrid cross? 2

14. Define heredity. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Mention any three traits that Mendel studied in pea plants. 3

16. How many types of gametes are produced by TT, Tt and tt? 3

17. What do you understand by the term 'independent assortment'? 3

OR

Differentiate between dominant and recessive traits. 3

18. What is the importance of variation to a species? 3

OR

What is a Punnett square? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the monohybrid cross between a tall and a dwarf pea plant. 5

OR

Explain the terms: allele, homozygous, heterozygous, F1, F2. 5

20. Describe the dihybrid cross of round yellow and wrinkled green seeds. 5

OR

How do chromosomes and genes control the inheritance of traits? 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND